

**Course:** Full Stack Web Development

**Course Code:** 23CM4121

**Task No.:** 1.1

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**Roll No.:** A24126552125

**Date:** 25-08-2026

## Title

Design Static Webpage Using HTML Components

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## Aim

To design and develop a **static webpage using HTML components** and understand the usage of various HTML elements such as headings, paragraphs, links, lists, tables, forms, audio, video, sections, and footer.

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## OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML webpage.
  - To learn and use different HTML elements.
  - To create headings and paragraphs using HTML.
  - To implement hyperlinks using the <a> element.
  - To display information using ordered and unordered lists.
  - To create and format tables using HTML table elements.
  - To design a student registration form using form controls.
  - To embed audio and video content in a webpage.
  - To understand the use of semantic elements such as <section> and <footer>.
  - To develop a complete static webpage using HTML components.
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## THEORY

**HTML (HyperText Markup Language)** is the standard markup language used to create and structure webpages. HTML uses different elements and tags to define the structure and content of a webpage.

In this experiment, various HTML components are used to create a static webpage containing information about **Anil Neerukonda Institute of Technology & Sciences (ANITS)**, student details, a registration form, and multimedia content.

The important HTML components used are:

- **<h1>, <h2>, <h3>** – Used to define headings of different levels.
- **<p>** – Used to display paragraphs.
- **<a>** – Used to create hyperlinks to external webpages.
- **<b>, <em>, <u>** – Used for bold, emphasized, and underlined text.
- **<ul> and <li>** – Used to create unordered lists.
- **<ol> and <li>** – Used to create ordered lists.
- **<table>** – Used to display information in rows and columns.
- **<form>** – Used to collect user input.
- **<input>** – Used to create text, email, password, radio button, and checkbox controls.
- **<select> and <option>** – Used to create a dropdown list.
- **<textarea>** – Used to accept multi-line text input.
- **<button>** – Used to create form buttons.
- **<audio>** – Used to embed audio content.
- **<video>** – Used to embed video content.
- **<section>** – Used to divide the webpage into logical sections.
- **<footer>** – Used to define the footer of the webpage.

The basic structure of the webpage can be represented as:

**HTML Document → Sections → Text/List/Table/Form → Multimedia → Footer**

Since the webpage is static, the content is directly written in HTML and does not require server-side processing or a database.

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## **ALGORITHM / PROCEDURE**

1. Create a new HTML file.
2. Define the basic HTML document structure.
3. Create an **About** section using the **<section>** element.
4. Add headings and paragraph content using heading and paragraph tags.

5. Add a hyperlink to the ANITS website using the <a> element.
6. Create an unordered list to display the courses offered.
7. Create another unordered list to display the departments.
8. Create an ordered list to display the application procedure.
9. Create a **Student Details** section.
10. Design a table using <table>, <tr>, <th>, and <td> elements.
11. Create a **Student Registration Form**.
12. Add text, email, password, radio button, checkbox, dropdown, textarea, and buttons to the form.
13. Embed an audio file using the <audio> element.
14. Embed a video file using the <video> element.
15. Add a footer using the <footer> element.
16. Save the HTML file.
17. Open the HTML file in a web browser.
18. Verify that all sections, lists, table, form, audio, video, and footer are displayed correctly.
19. Perform multiple test cases to verify the webpage components.
20. Compare the expected output with the actual output.

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## Program :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>
<body>
  <header>
    Anil Neerukonda Institute of Technology & Sciences (ANITS)</header>
    <nav>
      <a href="#">Home</a>
      <a href="#">About</a>
      <a href="#">Contact</a>
      <a href="#">Student Details</a>
      <a href="#">Student Registration Form</a>
    </nav>
    <section id="about">
      <h2>Welcome to Anits</h2>
      <p><a href="https://anits.org/">Anil Neerukonda Institute of Technology & Sciences (ANITS)</a>, Visakhapatnam is owned by <em>Megha Engineering and Infrastructure Limited (WEIL)</em> which is one of the top infrastructure and manufacturing company in India with headquarters at Hyderabad. The institute was established in the academic year <em>2001-02</em> by Anil Neerukonda Educational Society (ANES) which was founded by <b>Dr. N.B.R. Prasad</b>, an NRI Philanthropist from the USA, with industrialists and eminent educationists in memory of his son Late Anil Neerukonda.</p>
      <h3>Courses offered</h3>
      <ul>
        <li>B.Tech</li>
        <li>M.Tech</li>
        <li>MBA</li>
        <li>MCA</li>
      </ul>
      <h3>Departments</h3>
      <ul>
```

```
<h3>Departments</h3>
<ul>
  <li>Computer Science and Engineering</li>
  <li>Artificial Intelligence & Machine Learning</li>
  <li>Electronics and Communication Engineering</li>
  <li>Mechanical Engineering</li>
  <li>Civil Engineering</li>
</ul>
<h3>To Apply</h3>
<ol>
  <li>Register online.</li>
  <li>Fill in your personal details.</li>
  <li>Upload certificates.</li>
  <li>Pay the application fee.</li>
  <li>Download the confirmation receipt.</li>
</ol>
</section>

<section id="student-details">
  <h3>Student Details</h3>
  <table>
    <tr>
      <th>Roll No</th>
      <th>Name</th>
      <th>Branch</th>
      <th>Email</th>
    </tr>

    <tr>
      <td>A24126552125</td>
      <td>Vanisree</td>
      <td>Csm</td>
      <td>vanisreemesala.csm@anits.edu.in</td>
    </tr>

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    </tr>
  </table>
</section>
```

```
<section id="form">
  <h2>Student Registration <Form></Form></h2>
  <form action="">
    <label for="">Name : </label><input type="text" placeholder="Name"><br>
    <label for="">Email : </label><input type="email" placeholder="enter your mail"><br>
    <label for="">password : </label><input type="password" placeholder="enter your password"><br>
    <label for="">Female </label><input type="radio">
    <label for="">male </label><input type="radio">

    <p>Hobbies</p>
    <label for="">Reading Books</label><input type="checkbox" name="" id=""><br>
    <label for="">Watching TV</label><input type="checkbox" name="" id=""><br>
    <label for="">Coding</label><input type="checkbox" name="" id=""><br>
    <label for="">Playing</label><input type="checkbox" name="" id=""><br>

    <select id="branch" name="branch">
      <option>CSE</option>
      <option>AI & ML</option>
      <option>ECE</option>
      <option>EEE</option>
      <option>Mechanical</option>
      <option>Civil</option>
    </select>

    <br>
    <label for="">Comments</label><br>
    <textarea name="" id="" rows="4" cols="20"></textarea>
    <br>
    <button>Submit</button>
    <button>Reset</button>
  </form>
</section>

<audio src="" controls>
  <source src="./sigmamusicart-football-football-music-551346.mp3">
</audio>

<video src="" controls width="400">
  <source src="./367164_medium.mp4">
</video>

<footer>
  2026 @Task-1
</footer>
</body>
</html>
```

# Code explanation:

HTML (HyperText Markup Language) is used to create and structure the content of web pages. In this experiment, a static webpage is designed for **Anil Neerukonda Institute of Technology & Sciences (ANITS)** using different HTML components.

The webpage contains information about the college, courses offered, departments, and application procedure. It also includes a **student details table**, **student registration form**, **audio and video elements**, and a **footer**. Various HTML tags such as <section>, <h1>, <h2>, <p>, <a>, <ul>, <ol>, <table>, <form>, <input>, <select>, <textarea>, <button>, <audio>, <video>, and <footer> are used.

## HTML Components Used

| HTML Component   | Purpose                                     |
|------------------|---------------------------------------------|
| <section>        | Divides the webpage into different sections |
| <h1>, <h2>, <h3> | Displays headings                           |
| <p>              | Displays paragraphs                         |
| <a>              | Creates a hyperlink                         |

|                  |                                              |
|------------------|----------------------------------------------|
| <b>, <em>, <u>   | Provides text formatting                     |
| <ul>, <ol>, <li> | Creates unordered and ordered lists          |
| <table>          | Displays student information in tabular form |
| <form>           | Creates a student registration form          |
| <input>          | Accepts user input                           |
| <select>         | Provides a dropdown list                     |
| <textarea>       | Accepts multiple lines of text               |
| <button>         | Creates buttons                              |
| <audio>          | Embeds audio                                 |
| <video>          | Embeds video                                 |
| <footer>         | Displays footer information                  |

## TEST CASES AND EXECUTION DATA

The webpage was tested by checking each major HTML component and verifying whether it was displayed correctly in the browser.

| Test Case | Component Tested             | Action / Input                   | Expected Result                                           | Actual Result                | Status |
|-----------|------------------------------|----------------------------------|-----------------------------------------------------------|------------------------------|--------|
| TC-01     | Headings, Paragraphs & Lists | Open the webpage                 | Content, headings and lists should be displayed correctly | Displayed correctly          | Pass   |
| TC-02     | Hyperlink                    | Click the ANITS link             | ANITS website should open                                 | Website opened successfully  | Pass   |
| TC-03     | Student Details Table        | View the table                   | Student details should be displayed in rows and columns   | Table displayed correctly    | Pass   |
| TC-04     | Registration Form            | Enter details and select options | Form controls should accept the given inputs              | Inputs accepted successfully | Pass   |

